

Objective: You are a data scientist working for a telecom company. Your goal is to develop data-driven strategies to increase customer upgrades to higher-value bundles.

Problem Statement:

- The company offers tiered service bundles with varying data allowances, voice minutes, and features.
- You are tasked with building a predictive model to identify customers who are likely to benefit from upgrading to a more expensive bundle.
- This will help the company optimize its service offerings and target upgrade promotions more effectively.

Data:

- Observation Period: Past 3 months (M1, M2, M3)
- Prediction Period: Next 1 month (forward-looking)
- Event Rate: Upgrading to a higher bundle (currently 1%)
- Data Points:
 - Mobile data usage (MB) - "Data"
 - Outgoing voice minutes - "og"
 - Total revenue per customer - "rev"
 - Previous rate plan description
 - Region of most frequent usage - "Most_Used_Region"
 - Line activation date - "aon"
 - Upgrade event flag (migrated to Bundle2000 or not) - "post_migration_plan"

Requirements:

A. Data Exploration and Feature Engineering:

1. **Exploratory Data Analysis (EDA):**
 - Analyse the distribution of data points like data usage, voice minutes, and revenue to identify potential outliers and patterns.
 - Explore relationships between features using relevant visualizations and statistical tests.
 - Analyse distribution of the upgrade event across different customer segments.
 - Explore filtering or segmentation techniques to increase the predicted percentage of customers who will upgrade to 1.5%.
2. **Feature Engineering:**
 - Create new features by combining existing data points to enhance model performance.

B. Predictive Modelling:

- Build a machine learning model to predict the likelihood of customers upgrading to a higher-tier bundle.
- Select an appropriate algorithm to achieve the highest accuracy. Analyse the model's performance using:

- **Decile analysis:** This will show how well the model separates upgraders from non-upgraders across different decile distribution.
- **Confusion matrix:** This will summarize the model's performance by showing predicted vs. actual upgrade status.
- Utilize techniques like K-fold cross-validation to ensure model generalizability.
- Identify the top and most influential features in predicting upgrade behaviour.

C. Segmentation Strategies:

- Based on the insights generated from data exploration and model interpretation, segment customers into distinct groups with similar upgrade propensities
- Target upgrade promotions towards customer segments with higher predicted upgrade rates and personalized offers based on their specific needs. For example, high data users might be offered plans with increased data allowances, while infrequent callers might be presented with plans emphasizing other features.

D. Association Rule Learning (Optional):

- Explore using association rule learning to uncover customer usage patterns associated with upgrades. These insights can further guide targeting strategies.

Additional Information:

- The specific programming language is open for candidates to choose (e.g., Python, R).